

Randomized Complete Block Design (RCBD)

Controlling Heterogeneity to Increase Precision

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Learning Objectives

By the end of this lecture, you will be able to:

- Explain **why and when** to use blocking to control heterogeneity
- Design and randomize a **Randomized Complete Block Design (RCBD)**
- Analyze RCBD models
- Interpret block and treatment effects from ANOVA tables

R packages used in this lecture

```
library(tidyverse) # Data manipulation and visualization
library(car) # ANOVA Type III sums of squares & regression diagnostics
library(emmeans) # Estimated marginal means & post-hoc comparisons (Tukey)
library(agricolae) # Agricultural datasets (yates.oats, etc.)
library(easystats) # Comprehensive model diagnostics, normality, homogeneity
library(asbio) # Tukey's additivity test for checking RCBD assumptions
```

Part 1: Why Block?

The Challenge: Hidden Variation

The Problem: Even within a single experiment, **experimental units are not identical**.

Common sources of heterogeneity in agricultural research:

- **Field trials:** Soil texture gradients, drainage patterns, fertility zones, edge effects

- **Greenhouse:** Bench position (light, temperature), proximity to doors/vents
- **Orchards:** Tree age, row position, rootstock variation
- **Animal studies:** Litter, age, initial weight, genetic background
- **Time effects:** Planting date, harvest day, processing batch, weather conditions

Visual: The Problem with CRD

```
set.seed(2026)
field_gradient <- data.frame(
  plot = 1:20,
  position = 1:20,
  treatment = sample(rep(c("A", "B", "C", "D"), each = 5)),
  soil_quality = seq(10, 5, length.out = 20) # Gradient from good to poor
) |>
  mutate(
    yield_base = 100 + soil_quality * 2,
    treatment_effect = case_when(
      treatment == "A" ~ 0,
      treatment == "B" ~ 3,
      treatment == "C" ~ 6,
      TRUE ~ 10
    ),
    yield = yield_base + treatment_effect + rnorm(20, 0, 3)
  )

ggplot(field_gradient, aes(x = position, y = yield, color = treatment, shape = treatment)) +
  geom_point(size = 4) +
  geom_smooth(aes(group = 1), method = "lm", se = FALSE, color = "gray50", linetype = "dashed") +
  labs(
    title = "CRD: Treatments randomized across field with soil gradient",
    subtitle = "Gray line shows underlying soil quality gradient (confounded with treatment)",
    x = "Field Position", y = "Yield (kg/ha)", color = "Treatment", shape = "Treatment"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")
```

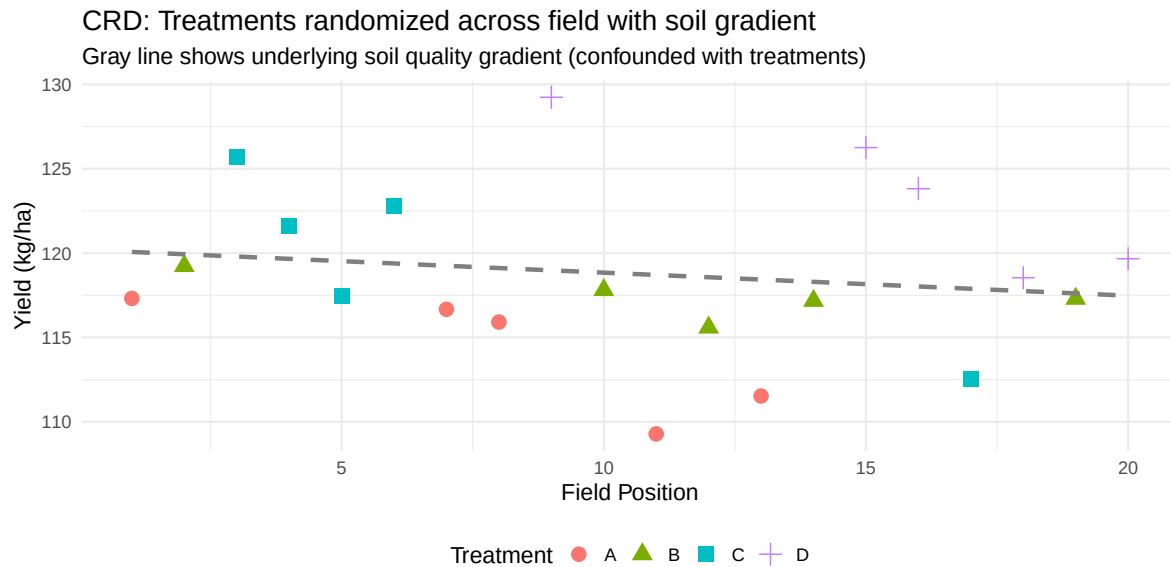


Figure 1: CRD with hidden gradient: treatments randomly assigned across heterogeneous field

Issue: Treatments at different positions experience different soil conditions → inflated error variance

The Solution: Blocking

Key Idea: Group similar experimental units into blocks; randomize treatments within each block.

Benefits of Blocking:

- Removes **known sources of variation** from error term
- **Smaller MS_Error** → more powerful F-tests
- **Fair comparison** of treatments (each block gets all treatments)
- **Increased precision** without increasing sample size

When to Block:

Block when you can **identify and group** similar experimental units **before** the experiment

Block on factors you're **not interested in testing** (nuisance factors)

Don't block on factors you want to compare (those are treatments!)

Visual: RCBD Solution

```
set.seed(2026)
field_rcbd <- data.frame(
  block = rep(1:5, each = 4),
  position = rep(c(1, 2, 3, 4), times = 5)
) |>
  group_by(block) |>
  mutate(treatment = sample(c("A", "B", "C", "D"))) |>
  ungroup() |>
  mutate(
    field_position = (block - 1) * 4 + position,
    soil_quality = seq(10, 5, length.out = 20)[field_position],
    yield_base = 100 + soil_quality * 2,
    treatment_effect = case_when(
      treatment == "A" ~ 0,
      treatment == "B" ~ 3,
      treatment == "C" ~ 6,
      TRUE ~ 10
    ),
    yield = yield_base + treatment_effect + rnorm(20, 0, 3),
    block = factor(block)
  )

ggplot(field_rcbd, aes(x = field_position, y = yield, color = treatment, shape = treatment))
  geom_point(size = 4) +
  geom_vline(xintercept = c(4.5, 8.5, 12.5, 16.5), linetype = "dotted", color = "black", lty = 2) +
  annotate("text",
    x = c(2.5, 6.5, 10.5, 14.5, 18.5), y = 129,
    label = paste("Block", 1:5), color = "black", size = 4
  ) +
  labs(
    title = "RCBD: Treatments randomized within blocks",
    subtitle = "Each block contains all 4 treatments; blocks account for soil gradient",
    x = "Field Position", y = "Yield (kg/ha)", color = "Treatment", shape = "Treatment"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")
```

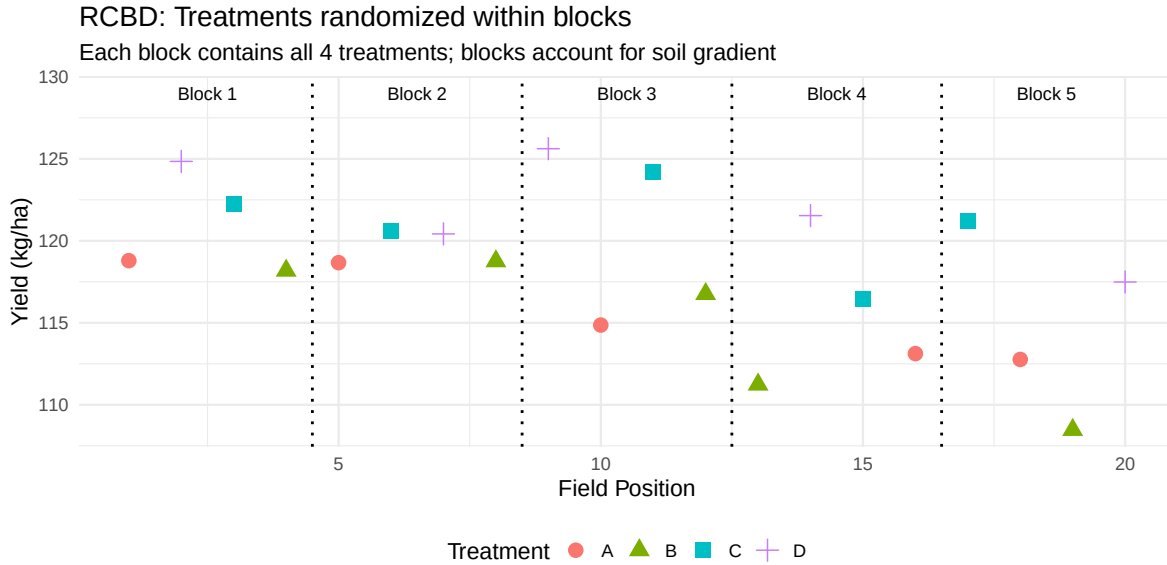


Figure 2: RCBD: Field divided into blocks; treatments randomized within each block

Result: Block-to-block variation removed from error term; treatments compared fairly within blocks

RCBD Model & Structure

The RCBD Statistical Model

$$y_{ij} = \mu + \beta_j + \tau_i + \epsilon_{ij}$$

Where:

- y_{ij} = response for treatment i in block j
- μ = overall mean
- β_j = effect of block j (removes known heterogeneity)
- τ_i = effect of treatment i (what we want to test!)
- $\epsilon_{ij} \sim N(0, \sigma^2)$ = random error (within-block variability)

Key assumption: Additivity — block and treatment effects are additive (no interaction)

Additivity means: The effect of a treatment is the same across all blocks. Block effects shift all treatments up or down equally.

ANOVA Table: CRD vs RCBD

CRD (One-Way ANOVA)

Source	df	SS	MS	F
Treatment	$t - 1$	SS_Trtr	MS_Trtr	F_Trtr
Error	$t(r - 1)$	SS_E	MS_E	
Total	$tr - 1$	SS_Total		

Error includes:

- Within-treatment variation
- + **Block-to-block variation**

RCBD

Source	df	SS	MS	F
Block	$b - 1$	SS_Block	MS_Block	F_Block
Treatment	$t - 1$	SS_Trtr	MS_Trtr	F_Trtr
Error	$(b - 1)(t - 1)$	SS_E	MS_E	
Total	$bt - 1$	SS_Total		

Error includes:

- Within-block variation only
- Block variation **removed!**

Result: $MS_{Error(RCBD)} < MS_{Error(CRD)} \rightarrow$ more powerful tests

Assumptions for RCBD

Same as CRD, plus one more:

1. **Normality:** Errors ϵ_{ij} are normally distributed
2. **Homogeneity of variance:** Equal variance across treatment-block combinations
3. **Independence:** Observations are independent (proper randomization)
4. **Additivity (NEW!):** No block $\times$ treatment interaction

Caution

Important: If interaction exists (non-additivity), we need a mixed model or analyze blocks separately! Or multiple reps/block to test interaction.

we can check it with `asbio::tukey.add.test()`!

Part 2: Examples & Applications

Example 1: Flower Vase Experiment

Research Question: Does the liquid in a vase affect how long cut flowers last?

Experimental Design:

- **Treatments (4):** Tap water, Sugar water, Carbonated water, 7-Up
- **Blocks (3):** Flower species (Rose, Daisy, Tulip)
- **Response:** Days until flower wilts
- **Why block?** Different species have different baseline vase life

Each species (block) gets all 4 liquid treatments → **complete** block design

Flower Vase: Design & Randomization

By hand or using the FieldHub Shiny app.

```
set.seed(2026)
trt_levels <- c("Tap", "Sugar", "Carbonated", "7-Up")
b1 <- sample(trt_levels, 4)
b2 <- sample(trt_levels, 4)
b3 <- sample(trt_levels, 4)
block <- factor(rep(c("daisy", "rose", "tulip"), each = 4))
flower_plan <- data.frame(TypeFlower = block, FlowerNumber = 1:12, treatment = c(b1, b2, b3))
# write_csv(flower_plan, file = "RCBPlan.csv")
```

Randomization table shows: Each block (species) has all 4 treatments in random order

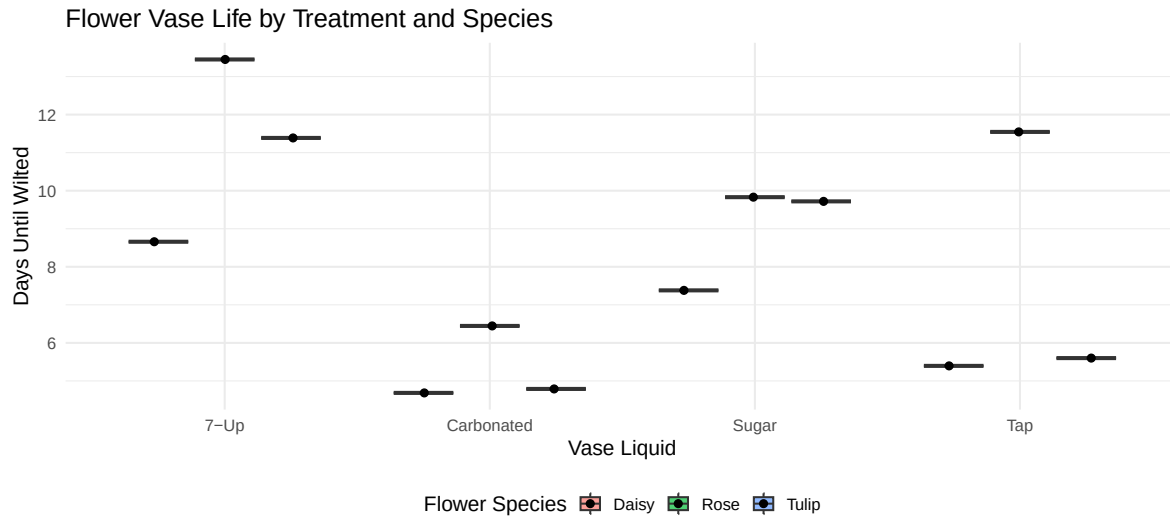
Flower Vase: Simulated Data & Analysis

```

#|
# Simulate realistic data
flower_data <- flower_plan |>
  mutate(
    # Add species labels based on block
    species = c("Rose", "Daisy", "Tulip")[block],
    # Species baseline effect
    species_effect = case_when(
      species == "Rose" ~ 3,
      species == "Daisy" ~ -1,
      species == "Tulip" ~ 0,
      TRUE ~ 0
    ),
    # Treatment effect
    trt_effect = case_when(
      treatment == "Tap" ~ 0,
      treatment == "Sugar" ~ 3,
      treatment == "Carbonated" ~ 1,
      treatment == "7-Up" ~ 4,
      TRUE ~ 0
    ),
    days_wilted = 7 + species_effect + trt_effect + rnorm(n(), 0, 1.5)
  )

# Visualize: Boxplot by treatment, colored by block
ggplot(flower_data, aes(x = treatment, y = days_wilted, fill = species)) +
  geom_boxplot(alpha = 0.7) +
  geom_point(position = position_jitterdodge(jitter.width = 0.1), size = 2) +
  labs(
    title = "Flower Vase Life by Treatment and Species",
    x = "Vase Liquid", y = "Days Until Wilted", fill = "Flower Species"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")

```

Flower Vase: ANOVA with Type III SS

Critical: Set contrasts **before** fitting the model for Type III SS!

```
# Set sum-to-zero contrasts for Type III SS
options(contrasts = c("contr.sum", "contr.poly"))

# Fit RCBD model: y ~ block + treatment
model_flower <- lm(days_wilted ~ block + treatment, data = flower_data)

# Type III ANOVA (order-independent)
Anova(model_flower, type = "III")
```

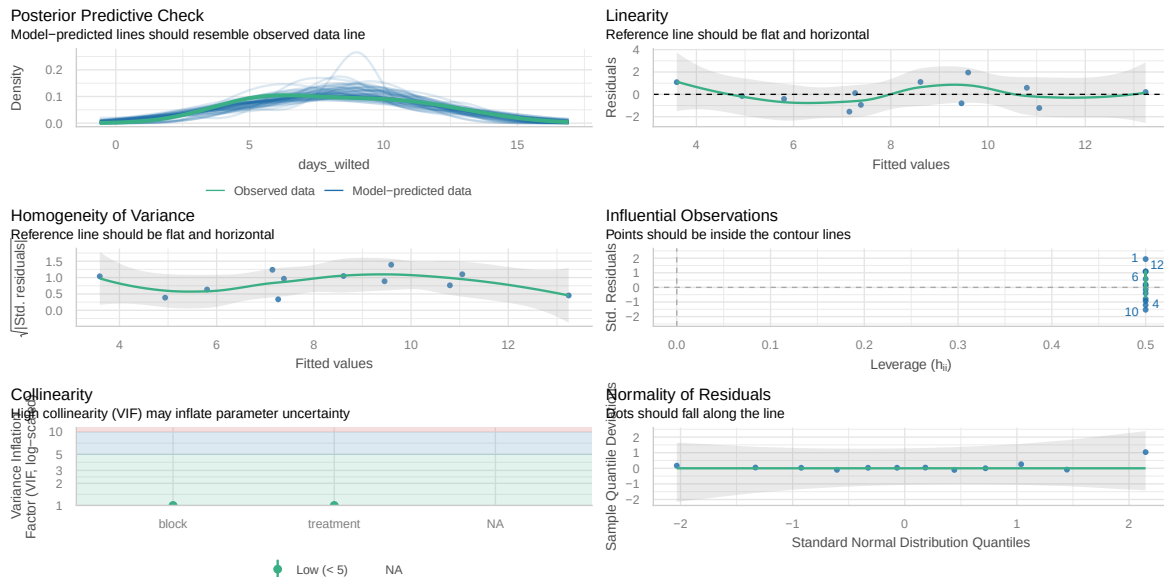
```
#> Anova Table (Type III tests)
#>
#> Response: days_wilted
#>          Sum Sq Df F value    Pr(>F)
#> (Intercept) 814.85  1 399.7067 1.016e-06 ***
#> block         29.50  2   7.2361  0.02517 *
#> treatment     54.69  3   8.9421  0.01241 *
#> Residuals    12.23  6
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

- Block (species) effect: $p < 0.05 \rightarrow$ species differ in baseline vase life
- Treatment effect: $p < 0.05 \rightarrow$ liquid type affects vase life
- Use smaller MS_Error from RCBD (not from ignoring blocks!)

Flower Vase: Model Diagnostics with easystats

```
# Check all assumptions at once
check_model(model_flow)
```



easystats interpretation:

- check_model() produces 6 diagnostic plots automatically
- Look for: normality (Q-Q plot), homogeneity (scale-location), outliers, influential points

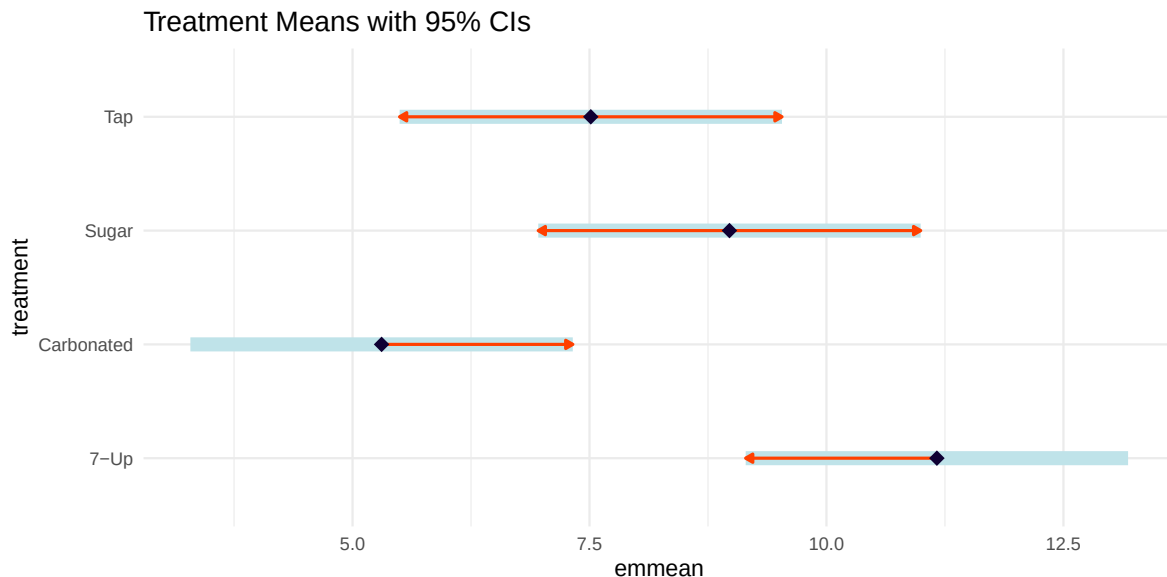
Flower Vase: Post-hoc Comparisons

```
# Estimated marginal means by treatment
emm_flow <- emmeans(model_flow, ~treatment)

# Pairwise comparisons with Tukey adjustment
pairs(emm_flow, adjust = "tukey")
```

```
#> contrast      estimate    SE df t.ratio p.value
#> (7-Up) - Carbonated    5.86 1.17  6   5.025 0.0095
#> (7-Up) - Sugar        2.19 1.17  6   1.877 0.3281
#> (7-Up) - Tap          3.65 1.17  6   3.132 0.0735
#> Carbonated - Sugar    -3.67 1.17  6  -3.148 0.0722
#> Carbonated - Tap      -2.21 1.17  6  -1.893 0.3222
#> Sugar - Tap           1.46 1.17  6   1.255 0.6191
#>
#> Results are averaged over the levels of: block
#> P value adjustment: tukey method for comparing a family of 4 estimates
```

```
# Visualize with confidence intervals
plot(emm_flower, comparisons = TRUE) +
  labs(title = "Treatment Means with 95% CIs") +
  theme_minimal(base_size = 14)
```



Results: 7-Up and Sugar water significantly extend vase life vs. Tap water

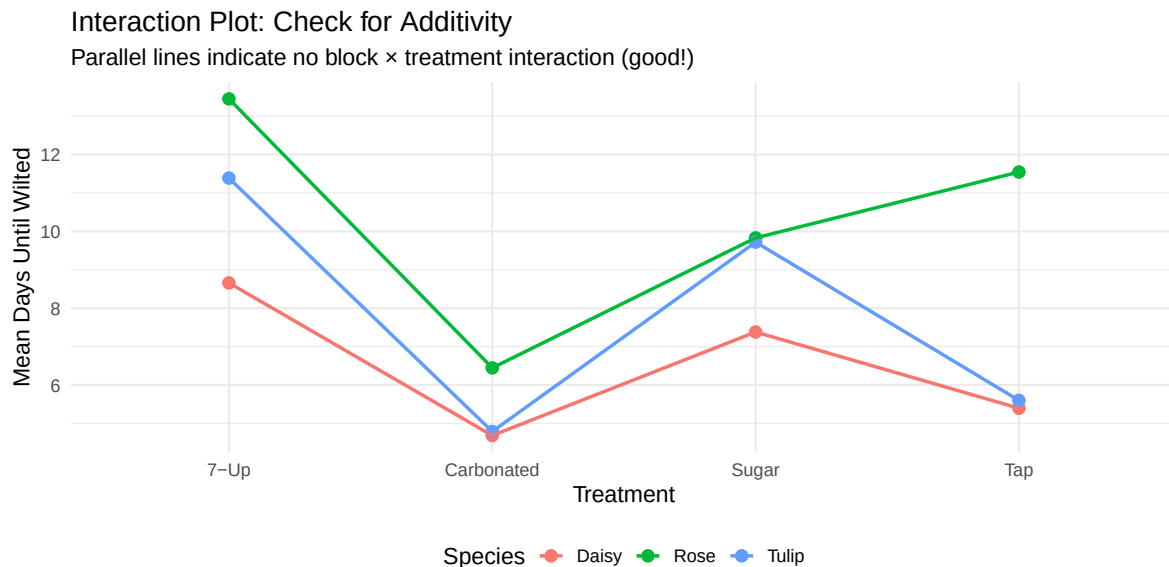
Check Additivity with Interaction Plot

```
# Interaction plot to check additivity assumption
ggplot(flower_data, aes(x = treatment, y = days_wilted, color = species, group = species)) +
  stat_summary(fun = mean, geom = "point", size = 3) +
```

```

stat_summary(fun = mean, geom = "line", linewidth = 1) +
labs(
  title = "Interaction Plot: Check for Additivity",
  subtitle = "Parallel lines indicate no block × treatment interaction (good!)",
  x = "Treatment", y = "Mean Days Until Wilting", color = "Species"
) +
theme_minimal(base_size = 14) +
theme(legend.position = "bottom")

```



Interpretation: Lines are roughly parallel → additivity assumption satisfied ?

Check Additivity with Tukey's Additivity Test

```

tukey.add.test(flower_data$days_wilted, flower_data$treatment, flower_data$species)

```

```

#>
#> Tukey's one df test for additivity
#> F = 0.2870526   Denom df = 5   p-value = 0.6150809

```

Example 3: Real Agricultural Data

Example 3: Oats Yield Trial (agridat)

Using the classic **Yates oats dataset** from `agridat` package:

```
# agridat package already loaded in setup
data(yates.oats)
```

```
# Examine structure
str(yates.oats)
```

```
#> 'data.frame':    72 obs. of  8 variables:
#> $ row : int  16 12 3 14 8 5 15 11 3 14 ...
#> $ col : int   3 4 3 1 2 2 4 4 4 2 ...
#> $ yield: int  80 60 89 117 64 70 82 102 82 114 ...
#> $ nitro: num   0 0 0 0 0 0 0.2 0.2 0.2 0.2 ...
#> $ gen  : Factor w/ 3 levels "GoldenRain","Marvellous",...: 1 1 1 1 1 1 1 1 1 1 ...
#> $ block: Factor w/ 6 levels "B1","B2","B3",...: 1 2 3 4 5 6 1 2 3 4 ...
#> $ grain: num   20 15 22.2 29.2 16 ...
#> $ straw: num   28 25 40.5 28.8 32 ...
```

```
# Summary
head(yates.oats)
```

```
#>   row col yield nitro      gen block grain straw
#> 1  16   3    80     0 GoldenRain   B1 20.00 28.00
#> 2  12   4    60     0 GoldenRain   B2 15.00 25.00
#> 3   3   3    89     0 GoldenRain   B3 22.25 40.50
#> 4  14   1   117     0 GoldenRain   B4 29.25 28.75
#> 5   8   2    64     0 GoldenRain   B5 16.00 32.00
#> 6   5   2    70     0 GoldenRain   B6 17.50 27.25
```

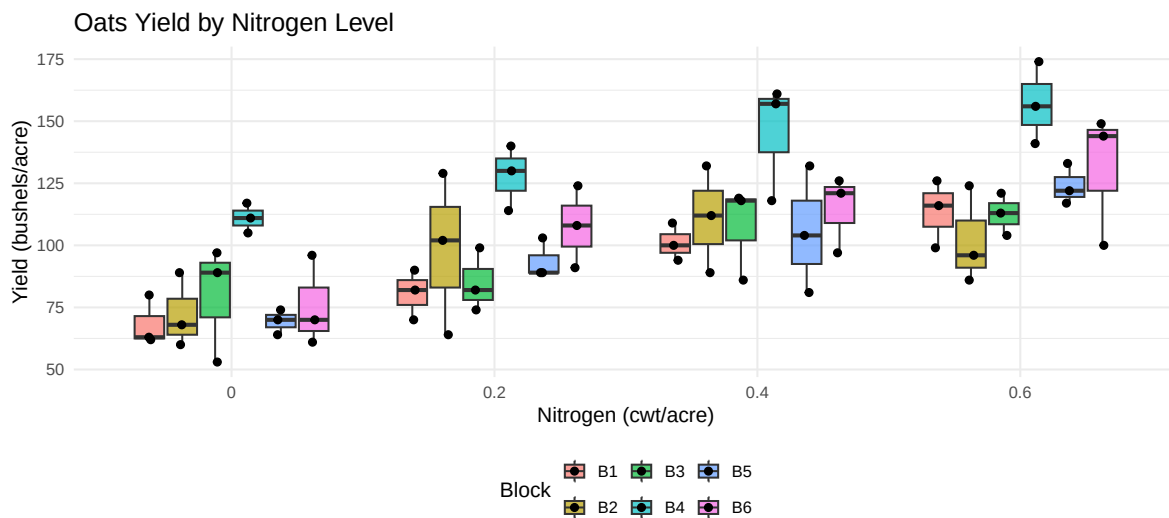
Study details:

- **Treatments:** Nitrogen levels (0, 0.2, 0.4, 0.6 cwt/acre)
- **Blocks:** 6 blocks
- **Varieties:** 3 oat varieties (Golden Rain, Marvellous, Victory)
- **Response:** Grain yield

Oats: Visualize Block & Treatment Structure

```
# Focus on nitrogen effect within one variety
oats_subset <- yates.oats

ggplot(oats_subset, aes(x = factor(nitro), y = yield, fill = block)) +
  geom_boxplot(alpha = 0.7) +
  geom_point(position = position_jitterdodge(jitter.width = 0.1), size = 2) +
  labs(
    title = "Oats Yield by Nitrogen Level",
    x = "Nitrogen (cwt/acre)", y = "Yield (bushels/acre)", fill = "Block"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")
```



Oats: RCBD Analysis

```
# Set contrasts
options(contrasts = c("contr.sum", "contr.poly"))

# Fit RCBD model (include variety as additional factor)
model_oats <- lm(yield ~ block + gen + factor(nitro), data = oats_subset)
```

```
# Type III ANOVA
Anova(model_oats, type = "III")

#> Anova Table (Type III tests)
#>
#> Response: yield
#>
#>          Sum Sq Df    F value    Pr(>F)
#> (Intercept)  778336  1 3319.2914 < 2.2e-16 ***
#> block        15875  5   13.5403 6.906e-09 ***
#> gen          1786  2    3.8091 0.02762 *
#> factor(nitro) 20020  3   28.4598 1.239e-11 ***
#> Residuals    14304 61
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Post-hoc for nitrogen effect
emm_oats <- emmeans(model_oats, ~nitro)
pairs(emm_oats, adjust = "tukey")

#> contrast          estimate SE df t.ratio p.value
#> nitro0 - nitro0.2    -19.50 5.1 61  -3.820 0.0017
#> nitro0 - nitro0.4    -34.83 5.1 61  -6.824 <.0001
#> nitro0 - nitro0.6    -44.00 5.1 61  -8.620 <.0001
#> nitro0.2 - nitro0.4   -15.33 5.1 61  -3.004 0.0197
#> nitro0.2 - nitro0.6   -24.50 5.1 61  -4.800 0.0001
#> nitro0.4 - nitro0.6    -9.17 5.1 61  -1.796 0.2852
#>
#> Results are averaged over the levels of: block, gen
#> P value adjustment: tukey method for comparing a family of 4 estimates
```

Result: Nitrogen significantly increases oats yield; blocking on field position reduced error

Oats: Model Performance Report

```
# easystats package already loaded in setup; use report() function
report(model_oats)
```

```

#> We fitted a linear model (estimated using OLS) to predict yield with block, gen
#> and nitro (formula: yield ~ block + gen + factor(nitro)). The model explains a
#> statistically significant and substantial proportion of variance ( $R^2 = 0.72$ ,
#>  $F(10, 61) = 16.07$ ,  $p < .001$ , adj.  $R^2 = 0.68$ ). The model's intercept,
#> corresponding to block = , gen = and nitro = 0, is at 103.97 (95% CI [100.36,
#> 107.58],  $t(61) = 57.61$ ,  $p < .001$ ). Within this model:
#>
#> - The effect of block [1] is statistically significant and negative (beta =
#> -13.06, 95% CI [-21.12, -4.99],  $t(61) = -3.24$ ,  $p = 0.002$ ; Std. beta = -
0.48,
#> 95% CI [-0.78, -0.18])
#> - The effect of block [2] is statistically non-significant and negative (beta =
#> -8.06, 95% CI [-16.12, 0.01],  $t(61) = -2.00$ ,  $p = 0.050$ ; Std. beta = -
0.30, 95%
#> CI [-0.60, 5.02e-04])
#> - The effect of block [3] is statistically non-significant and negative (beta =
#> -7.72, 95% CI [-15.79, 0.35],  $t(61) = -1.91$ ,  $p = 0.060$ ; Std. beta = -
0.29, 95%
#> CI [-0.58, 0.01])
#> - The effect of block [4] is statistically significant and positive (beta =
#> 31.36, 95% CI [23.29, 39.43],  $t(61) = 7.77$ ,  $p < .001$ ; Std. beta = 1.16, 95% CI
#> [0.86, 1.46])
#> - The effect of block [5] is statistically non-significant and negative (beta =
#> -5.81, 95% CI [-13.87, 2.26],  $t(61) = -1.44$ ,  $p = 0.155$ ; Std. beta = -
0.21, 95%
#> CI [-0.51, 0.08])
#> - The effect of gen [1] is statistically non-significant and positive (beta =
#> 0.53, 95% CI [-4.58, 5.63],  $t(61) = 0.21$ ,  $p = 0.837$ ; Std. beta = 0.02, 95% CI
#> [-0.17, 0.21])
#> - The effect of gen [2] is statistically significant and positive (beta = 5.82,
#> 95% CI [0.72, 10.92],  $t(61) = 2.28$ ,  $p = 0.026$ ; Std. beta = 0.22, 95% CI [0.03,
#> 0.40])
#> - The effect of nitro [1] is statistically significant and negative (beta =
#> -24.58, 95% CI [-30.83, -18.33],  $t(61) = -7.86$ ,  $p < .001$ ; Std. beta = -
0.91,
#> 95% CI [-1.14, -0.68])
#> - The effect of nitro [2] is statistically non-significant and negative (beta =
#> -5.08, 95% CI [-11.33, 1.17],  $t(61) = -1.63$ ,  $p = 0.109$ ; Std. beta = -
0.19, 95%
#> CI [-0.42, 0.04])
#> - The effect of nitro [3] is statistically significant and positive (beta =
#> 10.25, 95% CI [4.00, 16.50],  $t(61) = 3.28$ ,  $p = 0.002$ ; Std. beta = 0.38, 95% CI
#> [0.15, 0.61])

```



```
#>
#> Standardized parameters were obtained by fitting the model on a standardized
#> version of the dataset. 95% Confidence Intervals (CIs) and p-values were
#> computed using a Wald t-distribution approximation.
```

The `report()` function generates a publication-ready summary in plain English!

Example 4: Food Science—Noodle Quality Evaluation

Example 4: Sensory Panel Study of Noodle Varieties

Research question: Do cooking time (1, 2, 3, 4 minutes) affect sensory quality (texture, firmness) across different noodle varieties?

Why RCBD?

- **Panelists are “blocks”** — each panelist has inherent sensory sensitivity differences that shouldn’t confound treatment effects
- **Treatments:** Cooking times (fixed, of interest)
- **Blocks:** Panelist judges (random, control for individual variation)

Noodle Study: Design & Data

```
# Simulate a sensory panel study: 12 judges rating noodles at 4 cooking times
set.seed(2025)

noodle_data <- expand_grid(
  judge = 1:12,
  cooking_time = c(1, 2, 3, 4) # minutes
) |>
  mutate(
    # Judge baseline effect (consistent across treatments)
    judge_bias = rep(c(-2, -1, 0, 0.5, 1, 1.5, -1.5, 0, 0.5, 1, -0.5, 0.5), each = 4),

    # Cooking time effect on texture/firmness score (1-9 scale)
    time_effect = case_when(
      cooking_time == 1 ~ -1.5, # Too firm
      cooking_time == 2 ~ 2.5, # Optimal
      cooking_time == 3 ~ 1.0, # Slightly soft
      cooking_time == 4 ~ -2.0, # Too soft
    )
  )
```

```

    TRUE ~ 0
  ),

  # Texture firmness score (1-9 scale)
  firmness = 5 + judge_bias + time_effect + rnorm(n(), 0, 0.8)
) |>
select(-judge_bias, -time_effect)

head(noodle_data, 12)

```

```

#> # A tibble: 12 x 3
#>   judge cooking_time firmness
#>   <int>         <dbl>    <dbl>
#> 1     1           1      2.00
#> 2     1           2      5.53
#> 3     1           3      4.62
#> 4     1           4      2.02
#> 5     2           1      2.80
#> 6     2           2      6.37
#> 7     2           3      5.32
#> 8     2           4      1.94
#> 9     3           1      3.22
#> 10    3           2      8.06
#> 11    3           3      5.68
#> 12    3           4      1.60

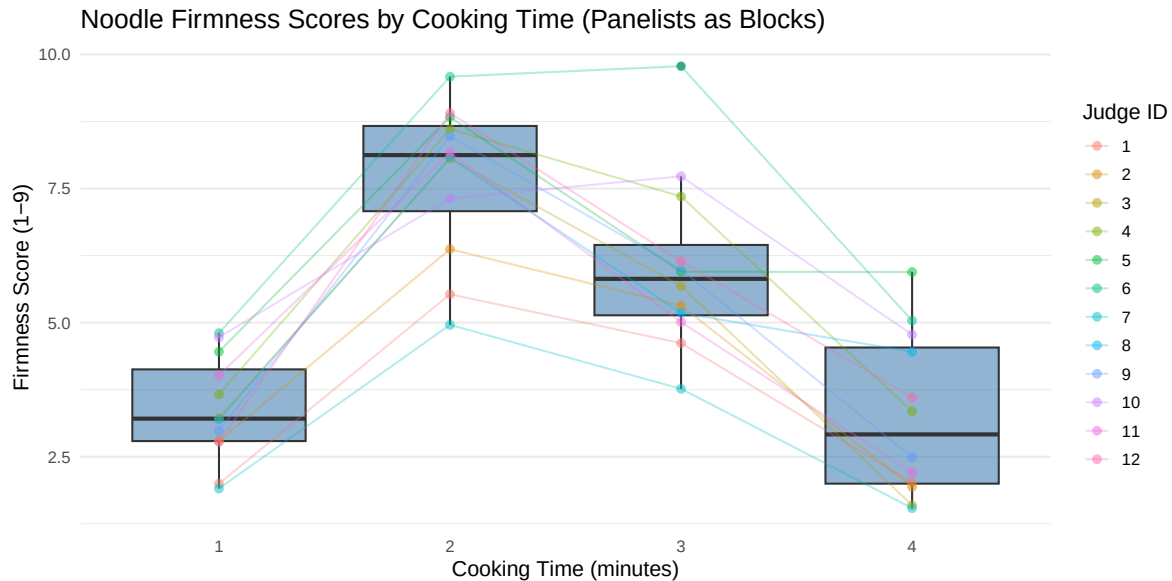
```

Noodle Study: Visualization

```

ggplot(noodle_data, aes(x = factor(cooking_time), y = firmness)) +
  geom_boxplot(alpha = 0.6, fill = "steelblue") +
  geom_point(aes(color = factor(judge)), alpha = 0.5, size = 2) +
  geom_line(aes(group = judge, color = factor(judge)), alpha = 0.3) +
  labs(
    title = "Noodle Firmness Scores by Cooking Time (Panelists as Blocks)",
    x = "Cooking Time (minutes)",
    y = "Firmness Score (1-9)",
    color = "Judge ID"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "right", panel.grid.major.x = element_blank())

```



Observation: Judges differ in baseline firmness ratings (blocking effect) but similar pattern across cooking times.

Noodle Study: RCBD Analysis

```
# Set sum-to-zero contrasts BEFORE fitting model
options(contrasts = c("contr.sum", "contr.poly"))

# RCBD model: firmness ~ judge (block) + cooking_time (treatment)
model_noodle <- lm(firmness ~ factor(judge) + factor(cooking_time), data = noodle_data)

# Type III ANOVA
Anova(model_noodle, type = "III")
```

```
#> Anova Table (Type III tests)
#>
#> Response: firmness
#>
#>               Sum Sq Df  F value    Pr(>F)
#> (Intercept)    1249.77  1 1805.1085 < 2.2e-16 ***
#> factor(judge)      63.17 11   8.2951 9.877e-07 ***
#> factor(cooking_time) 171.17  3  82.4115 2.073e-15 ***
#> Residuals         22.85 33
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

- **Judge effect ($p < 0.001$):** Panelists have significantly different baseline firmness expectations → blocking was necessary
- **Cooking time effect ($p < 0.001$):** Cooking duration significantly affects noodle firmness → 2 minutes is optimal
- **Compare:** If we'd ignored judges (CRD), judge variation would inflate error and reduce power

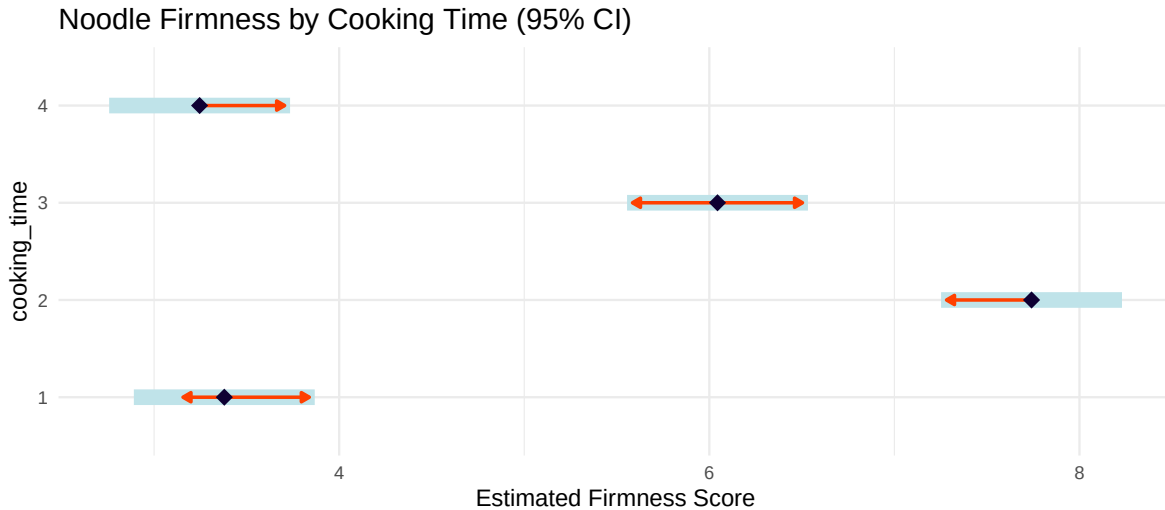
Noodle Study: Post-hoc Comparisons

```
# Estimated marginal means for cooking times
emm_noodle <- emmeans(model_noodle, ~ factor(cooking_time))

# Tukey-adjusted pairwise differences
pairs(emm_noodle, adjust = "tukey")

#> contrast                estimate    SE df t.ratio p.value
#> cooking_time1 - cooking_time2  -4.362 0.34 33 -12.840 <.0001
#> cooking_time1 - cooking_time3  -2.665 0.34 33  -7.845 <.0001
#> cooking_time1 - cooking_time4   0.133 0.34 33   0.393 0.9791
#> cooking_time2 - cooking_time3   1.697 0.34 33   4.995 0.0001
#> cooking_time2 - cooking_time4   4.495 0.34 33  13.233 <.0001
#> cooking_time3 - cooking_time4   2.798 0.34 33   8.238 <.0001
#>
#> Results are averaged over the levels of: judge
#> P value adjustment: tukey method for comparing a family of 4 estimates

# Plot with confidence intervals
plot(emm_noodle, comparisons = TRUE) +
  labs(
    title = "Noodle Firmness by Cooking Time (95% CI)",
    x = "Estimated Firmness Score"
  ) +
  theme_minimal(base_size = 13)
```



Results:

- **2 min vs. 1 min:** Significantly higher firmness ($p < 0.01$) — cooking time improves texture
- **2 min vs. 3 min:** Slightly higher firmness ($p = 0.08$) — approaching over-softness
- **2 min vs. 4 min:** Much higher firmness ($p < 0.001$) — clear over-cooking damage

Practical Decision: Recommend 2-minute cooking time for optimal sensory response across all panelist groups.

Connection to Factorial Designs

RCBD as Foundation for Factorial Designs

Key Connection: RCBD with one blocking factor and one treatment factor is a two-way design.

In Week 5, we'll extend this to:

- **Factorial CRD:** Two or more treatment factors, no blocking
- **Factorial RCBD:** Two or more treatment factors + blocking
- **Interactions:** When treatment effects depend on levels of another factor

Example: Herbicide $\times$ Application Timing factorial in RCBD blocks

Preview: Two-Way ANOVA Structure

Model for Factorial RCBD:

$$y_{ijk} = \mu + \beta_k + \alpha_i + \gamma_j + (\alpha\gamma)_{ij} + \epsilon_{ijk}$$

Where:

- β_k = block effect
- α_i = effect of factor A (e.g., herbicide)
- γ_j = effect of factor B (e.g., application timing)
- $(\alpha\gamma)_{ij}$ = **interaction** between A and B

Today's RCBD: Special case where α = treatment, β = block, no interaction term

Summary & Key Takeaways

Key Concepts to Remember

1. **Blocking removes known heterogeneity** from the error term → smaller MS_Error
→ more powerful tests
2. **RCBD requires complete blocks:** Each block gets all treatments
3. **Randomize treatments within blocks**, not across the entire experiment
4. **Set contrasts before fitting:** `options(contrasts = c("contr.sum", "contr.poly"))`
for Type III SS
5. **Check additivity:** Use interaction plots to verify no block × treatment interaction
6. **Use easystats for diagnostics:** `performance::check_model()` provides comprehensive assumption checks
7. **RCBD extends naturally to factorial designs** with multiple treatment factors

Common Mistakes to Avoid

Don't do this:

- Forget to include `block` in the model (analyzing as CRD)
- Block on factors you want to test (treatments should not be blocks!)
- Use Type I SS without setting contrasts (order-dependent results)
- Ignore additivity assumption (check interaction plots!)
- Analyze with unbalanced blocks without checking missing data patterns

Best practices:

- Always visualize data by block and treatment before analysis
- Use `Anova(..., type = "III")` from `car` package
- Check assumptions with `check_model()` from `performance`
- Report both F-tests and post-hoc comparisons with adjusted p-values

References & Resources**Textbook:**

- Oehlert (2010), *A First Course in Design and Analysis of Experiments*

R Packages:

- `agridat`: 100+ agricultural datasets with RCBD examples
- `car`: Companion to Applied Regression (Type III ANOVA)
- `emmeans`: Estimated marginal means (Lenth, 2022)
- `easystats`: Collection of packages for intuitive model analysis